

When the holdout lies: blind open-set recognition of 17,393 fish species on a thirty-submission budget

Lalith Sai · Cheng Yaw Low

Intelligent Systems Laboratory (ISLab), Changwon National University, Republic of Korea · System: FishONet v109 · github.com/lexus-x/FishONet

53.761% OVERALL **77.00%** PHOTOGRAPHED **23.77%** NO PHOTOGRAPH [official leaderboard · rank #3](#) · one uniform rule
no split membership used

ABSTRACT

Fish species recognition is simultaneously fine-grained, long-tailed, and open-set: closely related species differ by subtle morphology, the median photographed species has only two training images, and 11,598 of 17,393 candidate species have no training photograph. FishONet v109 addresses these challenges with one blind inference pipeline and reaches **53.761%** overall accuracy, ranking **#3** on the official leaderboard.

Each challenge receives a targeted solution. For fine-grained discrimination, BioCLIP encoders are adapted with additive-margin objectives from face recognition: ArcFace for the BioCLIP-2.5 members and CosFace (AM-Softmax) for the BioCLIP-2 adapter. For the long tail, frequency-aware Balanced Softmax is paired with class prototypes, nearest exemplars, and genus-level evidence so rare species are not represented by an unstable classifier head. For open-set recognition, we construct a multimodal support-set fish bank from taxonomy text, TreeOfLife-200M prototypes, and 117,225 disclosed iNaturalist photographs.

Our central implementation novelty is the **Proposed Quota Gate**: a learned 12-feature soft gate that ranks every evaluation image by familiarity and assigns the top 60% to the photographed-species route without reading seen/unseen membership. Leak-free route-specific re-ranking and genus backoff complete the system. Across 109 builds and thirty leaderboard submissions, the gate produced the largest real gain, while a machine-checkable audit confirms one uniform rule and one prediction over the full 17,393-class vocabulary.

Project links. Website: fishonet.lalithsai00.workers.dev · Code and artifacts: github.com/lexus-x/FishONet · Official results: codabench.org/competitions/16815/#/results-tab.

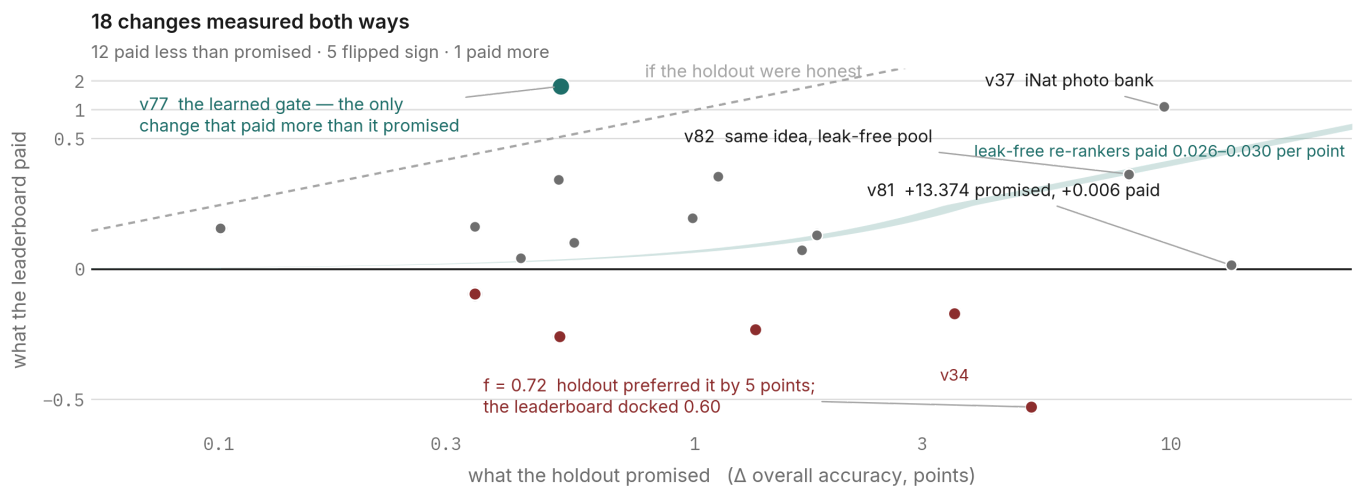


Figure 1. What the holdout promised, what the leaderboard paid. Every change for which both a holdout delta and a real leaderboard delta were recorded, in overall-accuracy points, symlog axes. The dashed diagonal is an honest proxy. Grey paid less than promised; red flipped sign; the one teal point, the learned gate, is the only change that paid more than it promised. v81 sits on the axis at far right: +13.374 promised, +0.006 paid. That is the leak of Section 5.1.

1 The problem

Each evaluation image receives one species from a fixed vocabulary of 17,393, scored on plain overall accuracy across 35,665 images. Because 20,097 (56.35%) belong to species seen in training and 15,568 (43.65%) do not, accuracy decomposes exactly as $0.5635 \cdot A_{\text{seen}} + 0.4365 \cdot A_{\text{novel}}$: a point on the novel side is worth 0.77 of a point on the seen side, and an image sent to the wrong side scores zero with certainty. Three difficulties compound.

Fine-grained. Sibling species often differ only in pigment pattern, stripe count, or fin morphology. We sharpen these small inter-class boundaries with additive-margin objectives developed for face recognition: ArcFace trains the BioCLIP-2.5 members, while a CosFace (AM-Softmax) objective adapts the BioCLIP-2 image encoder against image and text anchors.

Long-tailed. The 64,259 labelled images cover 5,795 species at a 140:1 imbalance ratio; the median species has two images, 79.4% have five or fewer, and 90.6% ten or fewer. Frequency-aware Balanced Softmax corrects the training prior, while inference combines class prototypes, nearest exemplars, and genus-level evidence instead of trusting an unstable learned classifier head for rare species.

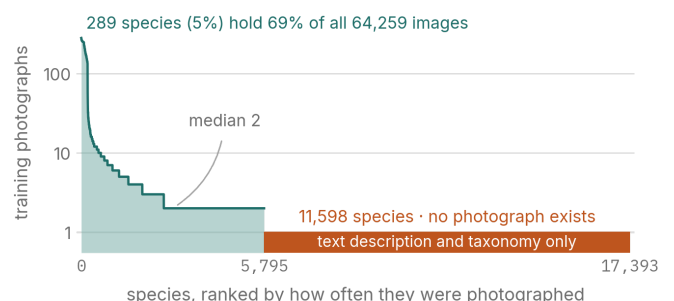


Figure 2. All three difficulties in one distribution. Training photographs per species, ranked, log scale, read directly from `label_train.json`. The teal curve is the long tail; it bottoms out at two images and stays there for half the photographed species. The clay bar is the open-set problem: 11,598 species for which no photograph exists at all.

Open-set. Two-thirds of the label space is unphotographed, so we build a multimodal support-set fish bank rather than reject these species as background. The novel route actively retrieves all 11,598 candidates using taxonomy and morphology text, TreeOfLife-200M prototypes, and disclosed iNaturalist

photographs embedded under complementary organismal encoders.

And a budget. Three submissions per day, thirty in total. The evaluation images are framed differently from the training ones

(mean aspect ratio 2.12 against 1.15), and every question about that distribution costs a submission. The offline holdout is the only other instrument we have. Figure 1 is what it turned out to be worth.

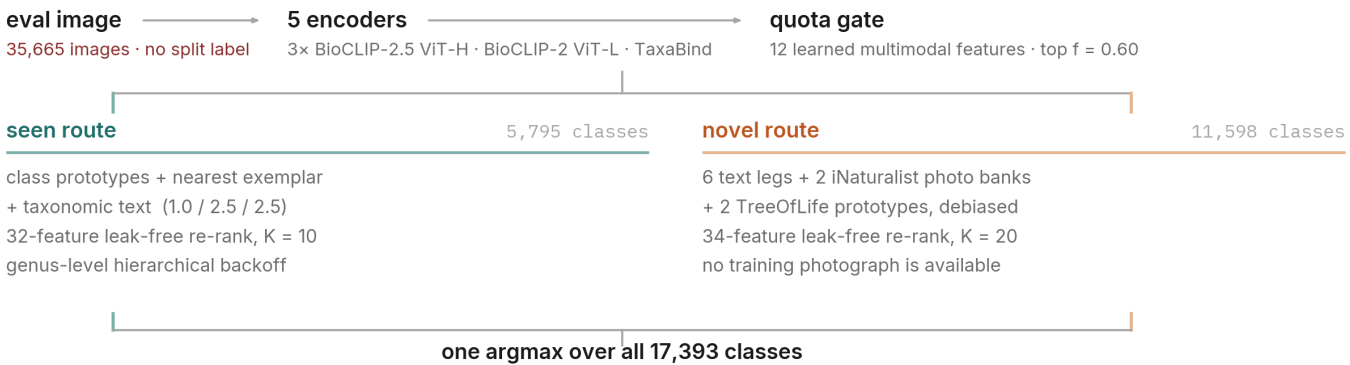


Figure 3. The v109 inference pipeline. One uniform rule over every evaluation image. The gate sees only signals computed from the image and the class texts — never split membership — and assigns a quota rather than thresholding. Both routes are scored and the label is one argmax over all 17,393 classes, so a prediction is never restricted to one side.

2 The rules, and how we comply

The competition forbids using the seen/unseen split at inference. `splits/test.pkl` and `splits/unseen.pkl` are protocol-only: a system may not read them to learn whether an evaluation image is of a photographed species, and may not restrict a prediction to one side of the split. The seen/novel decision must therefore be made *blind*, from the image and the class texts alone, and every error in it is unrecoverable. Section 6.2 shows this costs about 15.9% of all images and sets the design's ceiling.

Compliance is checked, not asserted. `research/verify_compliance.py` ships with the code, re-runs every claim in Table 1, and exits non-zero on any failure. Its central move is an inversion. It deliberately reads the split files, which no builder does, in order to prove the router could not have been an oracle: if we were consulting them, every seen-split image would reach a photographed class, and ours do so 89.11% of the time. The system makes the same errors a blind system makes.

Table 1. Every check re-run against the submitted artefact; all pass.

CLAIM	MACHINE-CHECKED EVIDENCE
No split file read at inference	static scan, 6/6 builders in the chain clean
One argmax over 17,393	21,399 predictions photographed, 14,266 unphotographed
Router is blind	89.11% / 77.58% route purity; an oracle scores 100 / 100
No fish-specific model	5 general organismal backbones; 2 fish fine-tunes declined
External data disclosed	iNat Open Data, TreeOfLife-200M; FishBase unused, declared
Gate artefact reproducible	<code>learned_gate_v79.pkl</code> rebuilds bit-identically

The one gray zone, priced instead of argued. Five points in the pipeline couple across the evaluation batch: member and gate standardisation, the f-quota rank cutoff, `dbnorm`'s across-image term, and Sinkhorn on the novel route. The rules forbid split-based routing and say nothing about batch-coupled inference either way. We did not want to argue the reading, so we built the strict alternatives and measured them. Freezing the two *routing* couplings against holdout-calibrated constants changes **729 of 35,665** predictions (2.04%); additionally freezing `dbnorm`'s hub term changes 4,213 (11.81%). Both builds ship. Organisers can pick a strictness level and re-run instead of accepting or rejecting the submission whole, and the licence grants them that right.

3 Problem-Aware Encoders and Data Resources

Training data is the provided 64,259 labelled images over 5,795 species, plus `descriptions.json` for all 17,393 classes; evaluation is the provided 35,665-image test set. Everything ran on one NVIDIA L40S (46 GB). Five encoders, all fine-tuned only on the provided images with shift-matched augmentation (`RandomResizedCrop, scale (0.35, 1.0), ratio (0.5, 2.0)`):

three BioCLIP-2.5 ViT-H/14 variants (`ctftshift`, contrastive LoRA with taxonomic context; `ftshift`; `fullft336shift` at 336 px), one BioCLIP-2 ViT-L/14 with a frozen backbone plus a LoRA adapter, and TaxaBind ViT-B/16 as a taxonomic anchor. No fish-specific model or dataset is used. External resources, fully disclosed: **iNaturalist Open Data** (117,225 research-grade images over 7,364 species, the novel route's photo bank) and **TreeOfLife-200M** embeddings in the BioCLIP-2 prototype legs.

4 Implementation

4.1 Seen and novel routes

For a candidate among the 5,795 photographed species the score is a weighted sum of mean class prototype, nearest training exemplar, and taxonomic text similarity, weights (1.0, 2.5, 2.5). The exemplar term dominates because of the long tail: at a median of two images per class a class mean is a poor summary, and one well-matched exemplar is the stronger evidence. For the 11,598 unphotographed species, ten legs are fused under a debiased normalisation (`dbnorm`) that removes each leg's per-class score bias: six text legs (multi-template taxonomy, common names, taxonomic context across four encoders), two iNaturalist photo-bank legs with 7-crop max-pooling, and two BioCLIP-2 prototype legs merged with TreeOfLife-200M.

4.2 Proposed Quota Gate

Routing is a ranking problem, not a thresholding problem. All 35,665 queries are scored by a 12-feature logistic model (seen-side maximum similarity, top-1/top-2 margin, log-sum-exp entropy gap, per-encoder maxima, the seen-text-minus-novel-text margin, and photo-bank confidence), and the top f = 0.60 quantile routes to the seen head. A quota is invariant to the scale drift the domain shift induces, and uniform by construction: the rule depends on nothing about an image but its rank.

4.3 Leak-free re-ranking, genus backoff

The top K candidates per route (10 seen, 20 novel) are re-ordered by a logistic re-ranker, 32 features seen-side and 34 novel-side. Every feature is a within-query statistic (percentile rank, gap-to-maximum, per-leg z-score), so it is invariant to candidate-pool size. Section 5.1 explains why that invariance is what separates a real gain from an artefact. The final build adds one seen-route signal: per candidate, `ctftshift` prototype and exemplar evidence pooled across sibling species of the same genus ($\lambda_G = 3.0$). It was submitted on 1 September, in the last slot before the 22:00 UTC close, changes 312 of 35,665 predictions (0.87%), and leaves the gate bit-identical.

5 What the holdout taught us

5.1 The leak, and what a proxy point is worth

Our first shortlist re-ranker (v81, submitted 29 August) scored **+13.374** on a class-disjoint 5-fold cross-validated pseudo-novel holdout and **+0.006** on the leaderboard. The cause was the pool, not the model. Gold labels in that holdout are always drawn from the 1,159 rarest *training* classes, which are only

9.1% of the 12,757-class candidate pool and which are identifiable from the features. So the re-ranker learned to prefer gold-eligible classes, picking one 53.7% of the time against the fusion's 38.4%, instead of learning to rank evidence. Class-disjoint cross-validation cannot catch this because the leak is in the pool's composition, not in the class partition. It took us one wasted submission to see it.

THE RULE WE NOW APPLY

Per-image decisions such as the gate are safe to train on the holdout. Per-candidate learning (re-rankers, learned fusion weights, calibrators over the class axis) must be trained on a pool in which every candidate is gold-eligible, with pool-size-invariant features. The magnitude of a holdout number is not evidence; the population it was measured on is.

Rebuilding the pool that way (v82) lowered the proxy gain to +8.154 while raising the real gain to +0.213. The seen head gave a consistent exchange rate, and that anchor, 0.026–0.030 real points per leak-free proxy point (the teal band in Figure 1), became a working instrument. A full point needs +34 proxy points, and nothing we had came close. Below about two proxy points the real outcome can flip sign (v84: +0.345 promised, -0.039 paid).

Table 2. The same idea, measured on a leaking pool and on a clean one.

RE-RANKER	CANDIDATE POOL	PROXY Δ	REAL Δ
v81 novel	leaky, 9.1% eligible	+13.374	+0.006
v82 novel	leak-free	+8.154	+0.213
v83 seen	leak-free	+1.803	+0.053

5.2 New evidence compounds, re-weighting refunds

The gate produced the project's largest single gain. It is the one teal point in Figure 1, the only change that paid more than it promised. Decomposing it twice gave us the most useful rule we found. A gate change moves overall accuracy through two channels: the *routing* term (how many images reach a route that can contain their label) and the *conditional* term (how accurate each route is on what it receives).

Table 3. Two changes to the same gate, decomposed. Real leaderboard deltas.

CHANGE TO THE GATE	ROUTING	COND.	REAL Δ
v56→v77 2 features → 12 new information	+1.082	+0.686	+1.767
v77→v79 same 12, C = 1 → 100 re-weighted only	+0.481	-0.439	+0.042

New features change *which* images sit near the boundary, and since they also predict whether the receiving head will be correct, both terms rise together. Re-weighting the same features only slides marginal images across a fixed ordering: routing improves, the conditional term falls almost as much, and 90% of the gain is refunded. We thereafter projected every gate proposal at 0.1× its routing delta and stopped tuning. The genus backoff was accepted on the same logic (sibling similarity is absent from the base fusion) and also transferred, which makes the rule 2 for 2.

5.3 The operating point, where the holdout is anti-correlated

The quota f trades the routes at roughly 4:1. The holdout preferred $f = 0.72$ by five points, and the leaderboard docked 0.60 for it. That is the red point at the bottom of Figure 1. Three real submissions then bracketed the axis and the response is concave, so $f = 0.60$ is a measured optimum: 53.520 / 53.697 / 53.492 at $f = 0.55$ / 0.60 / 0.65. The holdout was not consulted on this axis again.

5.4 Encoder Screening and Saturated Alternatives

Roughly 25 backbones were screened on the novel route. Organismal pre-training dominates: BioCLIP 21.53% where SigLIP2 reached 5.13% and BioTrove ≈0%. No encoder-side lever survived after v56, and every later real gain came from evidence the pipeline already computed. The following are closed and need not be re-run: **re-ranker model class** is inert (gradient boosting vs logistic on clean pools: -0.008 novel, +0.003 seen); **gate recalibration** is exhausted (six levers, AUC pinned at 0.981); **transduction** gives little (Sinkhorn +0.09 at v33, negative later); **multi-crop TTA** changed 0.6% of predictions; **recover-on-eject** promised +3.51 and paid -0.07.

6 Results

6.1 The campaign

Every number below is a real leaderboard score, not a holdout estimate. The system went from 45.39% to 53.761% over 109 numbered builds; the novel route carried almost all of it, 8.51% to 23.77%, while seen accuracy stayed inside a two-point band.

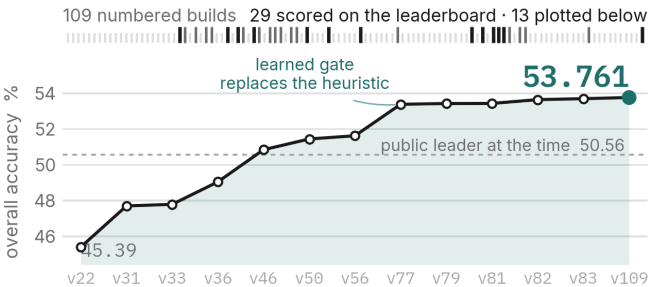


Figure 4. What 109 builds bought. Upper strip: every numbered build; the 29 that received a leaderboard score are darker, the 13 milestones plotted beneath darkest. A budget of 3 per day and 30 in total is why most builds were resolved offline. Lower: real overall accuracy at each plotted milestone.

Table 4. Measured leaderboard progression. Seen and novel columns are the organiser-reported diagnostics.

BUILD	SEEN	NOVEL	OVERALL	WHAT CHANGED
v22	—	—	45.390	closed-set shift-robust ensemble
v33	78.20	8.51	47.780	single-pipeline routing baseline
v56	75.67	20.57	51.616	7-crop max-pool, 336 px bank
v77	77.72	21.96	53.383	learned 12-feature gate
v82	77.45	22.91	53.644	leak-free novel re-ranker
v83	77.54	22.91	53.697	+ leak-free seen re-ranker
v109	77.00	23.77	53.761	+ genus-level backoff

6.2 Where the remaining error is

The gate separates the two populations almost perfectly on held-out data (AUC 0.981) but only at 0.911 on evaluation, because the framing shift moves the statistics of the crops it reads. At the deployed operating point it mis-routes 15.9% of images, each then choosing from a candidate set that cannot contain its label. Correcting only the routing would reach 54.35%, this design's ceiling. Reaching 55% needs +1.30 from the recognisers themselves, while re-ranking headroom is bounded at ~0.65 by the anchor of Section 5.1. The gap needs a stronger recogniser, not a better decision rule.

7 Three rules for thirty measurements

This benchmark forces fine-grained, long-tailed and open-set recognition at once, under a rule that denies the system any knowledge of which regime an image is in, and under a budget that makes the offline proxy the main instrument. The proxy lied to us more often than not. What we would carry to the next such problem is not the architecture but three rules for reading evidence when you get thirty real measurements.

- 1. Measure the population, not the number.** A holdout gain is only evidence if every candidate in the pool could have been the answer. Class-disjoint cross-validation does not check this. Ours said +13.374 and the leaderboard said +0.006.
- 2. New information compounds; re-weighting refunds.** Before spending a submission on a change to a decision rule, ask whether it reads evidence the rule has never seen. If not, project it at a tenth of its holdout delta.
- 3. When a rule is ambiguous, price the strict reading and ship it.** Building the per-image variant cost us nothing on the leaderboard and turned an argument into a number, 729 predictions. The decision then belongs to the organisers, which is where it should be.

REFERENCES [1] Stevens et al. BioCLIP. CVPR 2024. [2] Gu et al. BioCLIP 2 / TreeOfLife-200M. 2025. [3] Sastry et al. TaxaBind. WACV 2025. [4] Radford et al. CLIP. ICML 2021. [5] Tschannen et al. SigLIP 2. 2025. [6] Hu et al. LoRA. ICLR 2022. [7] Cuturi. Sinkhorn Distances. NeurIPS 2013. [8] iNaturalist Open Data, 2026.